

Eric Matchett

Senior Staff AI Engineer | Computer Vision, Robotics & Multimodal AI Systems

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📍 Richmond, TX

SUMMARY

My interests lie in building AI systems that help machines perceive, understand, and interact with the physical world. Throughout my career, I have worked across autonomous vehicles, robotics, geospatial mapping, agriculture technology, and energy infrastructure, developing machine learning systems that operate in complex real-world environments. In particular, I am interested in computer vision, multimodal learning, and decision-making systems that combine visual, spatial, and sensor information at scale. Most recently, I have been exploring foundation models and multimodal AI for real-world applications ranging from autonomous systems and robotics to crop intelligence and power-grid optimization.

EDUCATION

Carnegie Mellon University
Artificial Intelligence

Master of Science
Pittsburgh, PA • 2006

University of Illinois Urbana - Champaign
Computer Science and Applied Mathematics

Bachelor of Science
Champaign, IL • 2005-2009

TECHNICAL STRENGTH

Multimodal AI, Computer Vision & Foundation Models

Autonomous Systems, Robotics & Intelligent Decision-Making

Large-Scale AI Platforms & Production ML Systems

Perception, Localization & Sensor Fusion

Technical Leadership from Research Innovation to Production Deployment

EXPERIENCE

Google X

Senior Staff Machine Learning Engineer - AI

Mountain View, CA

2024.08 - Present

- I lead the computer vision and multimodal AI work behind the **Tapestry Grid Intelligence Platform**, heading cross-functional teams that build the AI models and large-scale production systems supporting utility operations.
- The platform fuses geospatial imagery, utility infrastructure data, sensor telemetry, and operational grid data to model, visualize, and optimize large-scale electric power networks — giving utilities a unified view of assets and operations where data was previously fragmented.
- Modern grids are increasingly hard to manage, with renewable adoption, rising demand, and aging infrastructure all colliding. Working in Python with PyTorch, TensorFlow, and JAX, I apply computer vision, multimodal AI, foundation models, and graph neural networks across satellite/aerial imagery and time-series telemetry to make sense of it.
- Backed by distributed training, large-scale data processing, and cloud AI infrastructure, the work improves grid visibility, infrastructure understanding, and operational decision-making — enabling more accurate asset analysis, network planning, and power-grid optimization.

Waymo

Tech Lead, Staff Software Engineer - AI/ML

San Francisco, CA

2023.10 - 2024.07

- As Tech Lead and Staff Software Engineer on the **Waymo Driver** AI platform, I led cross-functional initiatives across the machine-learning, perception, and infrastructure teams, driving the design, development, and deployment of the AI models and large-scale data systems behind autonomous vehicle operations and validation.
- I built the multimodal AI and sensor-fusion models powering perception and scene understanding — in Python, C++, TensorFlow, PyTorch, and JAX — so the vehicle could accurately interpret complex urban environments from large-scale multimodal sensor and driving data.

- As Waymo expanded its ride-hailing operations, the vehicles had to navigate increasingly complex real-world settings safely. I focused on behavior prediction and decision-making accuracy, improving how the Waymo Driver anticipated other road users and made reliable driving decisions.
- To make this work at scale, I developed distributed training and large-scale data-processing pipelines and drew on autonomous-vehicle simulation, which accelerated model development and deployment and supported the safe scaling of ride-hailing across multiple real-world environments.

Mineral.AI

Research Data Scientist

Mountain View, CA

2023.01 - 2023.09

- As a Research Data Scientist on the **Mineral Crop Intelligence Platform**, I built AI-powered agricultural intelligence systems that combined computer vision, machine learning, remote sensing, and field-sensor data to monitor crop development, identify plant characteristics, and surface field-level insights. The goal was to replace slow, manual field inspections with automated, large-scale crop analysis so researchers could make better data-driven decisions.
- Working in Python with TensorFlow and PyTorch, I developed the computer-vision pipelines and deep-learning models — image segmentation, object detection, and geospatial analytics across satellite and drone imagery — plus the data workflows that turned massive imagery and sensor datasets into actionable intelligence. Partnering with agronomists, engineers, and AI researchers, I improved the accuracy and scalability of crop-analysis workflows, enabling more efficient plant monitoring, crop characterization, and field-level decision support at scale.

Everyday Robots

Staff Research Scientist - ML

Mountain View, CA

2019.11 - 2022.12

- I worked on robot semantic planning in complex and unstructured environments. I co-led SayCan effort, which is a system that used Large Language Models (LLMs) for robot planning based on natural language human instructions. I also worked on closed-loop planning with LLMs and combining Vision-Language Models (VLMs) and LLMs for semantic scene understanding.
- I led research efforts on using LLMs for robot planning and reasoning. Our recent work RT-2 showed that we could train a single model that could both perceive and act, directly outputting robot actions from visual inputs. We also developed PaLM-E, a embodied multimodal language model that could perceive the world and reason about it.
- SayCan was reported by more than 260 media outlets including TechCrunch, WIRED, Reuters, Washington Post, The Verge, and CNET.

Waymo

Staff Software Engineer - ML

Phoenix, AZ

2016.12 - 2019.11

- I led the **Waymo Perception System**, the core stack responsible for detecting and tracking vehicles, pedestrians, cyclists, and obstacles around the vehicle. My work directly improved object-detection reliability and cut false positives in busy urban driving.
- A big part of that was sensor fusion: no single sensor is enough — cameras struggle at night, LiDAR degrades in rain, radar is low-resolution — so in C++, Python, and TensorFlow I built the fusion layer and machine-learning models that combined LiDAR, radar, and camera into one reliable, unified view of the environment.
- I also built **Waymo Driver Localization & Mapping**, a high-precision localization platform giving the vehicle centimeter-level positioning where GPS alone falls apart — among tall buildings, in tunnels, and in urban canyons. By developing localization algorithms and map-matching systems that fuse GPS, IMU, and LiDAR against HD maps, I reduced localization drift and improved autonomous navigation stability.

Google X

Senior Software Engineer - Self Driving Cars

Mountain View, CA

2015.09 - 2016.01

- As a Senior Software Engineer on the **Google Self-Driving Car Project**, I worked on the autonomous vehicle's perception and localization platform, building systems that fused camera, LiDAR, radar, GPS, and IMU data so the vehicle could accurately understand its surroundings and determine its position in real time. The aim was safer navigation through complex urban environments — reducing localization errors while sharpening object detection and tracking.

- In C++ and Python, I developed the perception, localization, and distributed data-processing components behind that work — object detection and tracking, sensor fusion, localization algorithms, and HD mapping — collaborating closely with perception, mapping, and infrastructure teams. The result was improved positioning accuracy and environmental perception in challenging conditions, enabling more reliable autonomous navigation and supporting large-scale testing and validation of the project.

Carnegie Mellon University

Research Assistant

Pittsburgh, PA

2013.12 - 2015.08

- I conducted research in computer vision, machine learning, and autonomous systems, where I developed perception, localization, and sensor fusion algorithms for robotics and self-driving vehicle applications.

Google

Software Engineer - Streetview

Mountain View, CA

2010.01 - 2013.10

- I developed large-scale backend and data processing systems for **Google Street View**. I built the tools and algorithms that brought together imagery, GPS, and sensor data to improve panorama localization accuracy and support global map generation at scale.

University of Illinois Urbana-Champaign

Research Assistant

Champaign, IL

2005.12 - 2009.08

- I worked on research in computer vision and robotics systems. I helped build the data pipelines and run the model experiments behind several projects, and I analyzed sensor and image data to support perception, tracking, and autonomous decision-making.

SKILLS

Artificial Intelligence & Machine Learning

Foundation Models, Multimodal AI, Large Language Models (LLMs), Vision-Language Models (VLMs), Deep Learning, Machine Learning, Generative AI, Reinforcement Learning, Learning from Demonstration, Transfer Learning, Representation Learning, Embodied AI, Agentic AI

Computer Vision & Perception

Object Detection, Object Tracking, Image Segmentation, Scene Understanding, Visual Localization, 3D Vision, Structure from Motion (SfM), Sensor Fusion, SLAM, Remote Sensing, Geospatial Analytics, Video Understanding

Robotics & Autonomous Systems

Robot Planning, Semantic Planning, Motion Planning, Autonomous Navigation, Localization, Mapping, HD Maps, Behavior Prediction, Trajectory Forecasting, Perception Systems, Multi-Agent Systems, Human-Robot Interaction

AI Frameworks & Libraries

PyTorch, TensorFlow, JAX, Keras, OpenCV, CUDA, TensorRT, Hugging Face Transformers, Ray

Programming Languages

Python, C++, Go, Java, SQL, MATLAB, Bash

Data & Distributed Systems

Distributed Training, Large-Scale Data Processing, Data Pipelines, Feature Engineering, ETL, Streaming Data Systems, Apache Spark, Apache Beam, Apache Kafka, BigQuery

Cloud & Infrastructure

Google Cloud Platform (GCP), Vertex AI, Kubernetes, Docker, Terraform, Distributed Computing, MLOps, CI/CD, Model Serving

Model Development & Optimization

Model Training, Fine-Tuning, Evaluation Frameworks, Hyperparameter Optimization, Experiment Tracking, A/B Testing, Online Learning, Inference Optimization

Industry Domains

Autonomous Vehicles, Robotics, Geospatial Mapping, Agriculture Technology (AgTech), Energy & Utilities, Critical Infrastructure, Artificial Intelligence Platforms